AGENTIC AI ENGINEERING

ASSIGNMENT 2

Safe and Production-Aware Student Support Agent

PGDM (Big Data Analytics / Artificial Intelligence & Data Science)
Batch 2025–27

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1. Executive Summary

Assignment 2 extends the same Student Support Agent developed in Assignment 1. The objective is to make the existing chat-style application safer and more suitable for institutional use by adding student identity, privacy enforcement, code-level authorization, approval for sensitive actions, audit logging, and continued tracing/evaluation.

The submitted project ZIP was inspected. Its audit log contains 20 events: 11 DENIED, 1 ALLOWED, 4 PENDING_APPROVAL, 2 REJECTED_NOT_EXECUTED, and 2 APPROVED_AND_EXECUTED. The approval-request file contains three requests. The local agent_traces.json file in the submitted ZIP is empty; therefore, Assignment 2 latency and token usage cannot be calculated from that file.

The developer reports that the required functional tests were completed successfully. Where the submitted files provide direct evidence, this report identifies the relevant audit records. Where evidence is not present, the report labels it as developer-reported or pending rather than inventing measurements.

2. Assignment Requirements and Implementation

Requirement in brief	Implementation in project
Student identity	The application maintains the currently signed-in student identity and uses it for student-specific operations.
Data privacy	Students may access their own private records; requests for another student's private data are blocked.
Code-level authorization	Authorization is enforced in Python application/tool code, not only through an LLM instruction.
Sensitive-action approval	Mobile-number and email-address changes are created as pending requests and require explicit student confirmation before execution.
Auditability	Important actions record actor, target, attempted action, status/decision, and result.
Tracing and evaluation	LangSmith tracing is retained from Assignment 1. Local trace file is present but empty in the supplied ZIP; attach LangSmith export/screenshots as separate evidence.

3. System Overview and Architecture

The project retains the Assignment 1 chat-style Streamlit application. Streamlit provides the user interface and session identity; the Groq-hosted model (openai/gpt-oss-120b) performs agent reasoning and tool selection; Python functions execute permitted operations; local JSON files store demonstration records and workflow state; BM25-style retrieval provides relevant knowledge context; and LangSmith is retained for tracing.

Student login → Streamlit chat → Agent / Groq model → Python authorization checks → permitted tool or approval workflow → JSON update only after confirmation → chat response.

Main application files include app.py, support_tools.py, retrieval.py, and knowledge_ingestion.py. The data directory contains student records, persistent memory, approval requests, audit logs, chat history, support requests, agent traces, and knowledge JSON files.

4. Privacy and Authorization Rules

- The authenticated student ID is the authority for student-specific data access.
- A tool request is checked against the authenticated identity before private data is returned or modified.
- A request targeting another student's private record is denied, even if the prompt asks the AI agent to retrieve it.
- Authorization is implemented in normal Python application/tool code; the LLM prompt is not treated as the security boundary.
- A denied request must not change the target student's record.

The supplied audit log includes a cross-student denial: actor AIDTM001 attempted to read the fee record for AIDTM002; the event status is DENIED and the reason is cross_student_access.

5. Sensitive Contact-Detail Change Workflow

Mobile-number and email-address changes use an approval workflow. The student submits a proposed value; the application creates a pending request without immediately changing the student record; an in-chat confirmation card presents the action; and the student can confirm or reject it.

On confirmation, the backend checks request ownership and executes the update for the authenticated student's record. On rejection, the request is marked REJECTED_NOT_EXECUTED and the original contact value remains unchanged. The application also displays a short success or rejection message after the action.

The submitted approval_requests.json contains three request records: two APPROVED_EXECUTED requests with executed=true and one REJECTED request with executed=false.

6. On-Demand Contact Details

The intended user experience is that the chatbot displays the signed-in student's contact details only when the student asks for them, such as “Show my contact details.” It must not display the details automatically in a fixed page section. A request for another student's contact details must be denied by backend authorization. This on-demand behaviour was added as a subsequent feature patch; the original submitted ZIP does not contain evidence of its final patched runtime, so include a fresh screenshot/video frame in the final evidence folder.

7. Persistent Memory Safety

The agent may retain useful non-sensitive facts, but OTPs and passwords provided in chat must not become persistent memory. The submitted student_memory.json contains only a programme fact for AIDTM001 and no visible OTP/password value. The developer also reports that the required secret-handling test passed. A dedicated test screenshot or trace should be attached to demonstrate the attempted secret and the resulting memory exclusion.

8. Audit Logging and Traceability

The supplied audit log contains 20 events. Status counts are: DENIED=11, ALLOWED=1, PENDING_APPROVAL=4, REJECTED_NOT_EXECUTED=2, APPROVED_AND_EXECUTED=2. These records provide evidence for own-record access, cross-student denial, pending contact updates, rejected requests, and approved execution.

The supplied data/agent_traces.json file is empty. LangSmith integration is described as retained, but no Assignment 2 trace export or numeric token/latency record was included

in the supplied ZIP. Do not treat the Assignment 1 metrics below as Assignment 2 measurements.

9. Sample Audit Evidence

Audit ID	Action	Actor	Target	Status	Relevant details
AUD-5942FD6121	read_fee_record	AIDTM001	AIDTM001	ALLOWED	Own fee record read
AUD-00ACD71F90	read_fee_record	AIDTM001	AIDTM002	DENIED	cross_student_access
AUD-A3F9738EEC	request_contact_update	AIDTM001	AIDTM001	PENDING_APPROVAL	mobile_number; request APR-321E60553F
AUD-A012223599	decide_contact_update	AIDTM001	AIDTM001	REJECTED_NOT_EXECUTED	mobile_number; request APR-321E60553F
AUD-D1F3C37678	execute_contact_update	AIDTM001	AIDTM001	APPROVED_AND_EXECUTED	mobile_number; changed=true
AUD-56E672FE2B	decide_contact_update	AIDTM001	AIDTM001	REJECTED_NOT_EXECUTED	email; request APR-51EC89C89B
AUD-62BE5203EA	execute_contact_update	AIDTM001	AIDTM001	APPROVED_AND_EXECUTED	mobile_number; changed=true

10. Final Evaluation Table – Assignment 1 vs Assignment 2

The assignment brief asks for task success, privacy tests, unsafe-action tests, latency, token usage, and trace/audit evidence. The following table separates the Assignment 1 sample baseline from Assignment 2 results. Assignment 1 values are individual sample traces, not averages.

Measure	Assignment 1 baseline	Assignment 2 result	Evidence / note
Normal task success	Working fee/policy/memory support tasks	PASS (developer-reported); own fee access also present in audit log	A2 audit AUD-5942FD6121: ALLOWED
Privacy tests	No equivalent code-level privacy control documented in A1 baseline	PASS (developer-reported); cross-student fee read blocked	A2 audit AUD-00ACD71F90: DENIED
Sensitive action approval	No approval-gated contact update in A1 baseline	PASS; pending, rejected and approved-executed states recorded	4 pending, 2 rejected-not-executed, 2 approved-and-executed events
Latency	1.914–2.719 s across four individual sample traces	Not measurable from supplied A2 trace file	agent_traces.json is empty; capture LangSmith run values
Input tokens	1,115–4,304 across four sample traces	Not available in supplied A2 evidence	Capture LangSmith input-token values
Output tokens	127–272 across four sample traces	Not available in supplied A2 evidence	Capture LangSmith output-token values

Trace / audit evidence	LangSmith sample traces documented in A1	Audit evidence present; A2 LangSmith export pending	Attach LangSmith screenshots/export for final submission

11. Assignment 1 Baseline Trace Samples

A1 sample task	Latency	Prompt/input tokens	Completion/output tokens	Note
Fee-status trace	1.914 s	3,021	144	Individual Assignment 1 sample trace
Payment-policy trace	2.421 s	4,304	272	Individual Assignment 1 sample trace
Memory trace	~2.287 s	~1,115	~127	Approximate values from Assignment 1 screenshot
General AI/ML/DL answer	2.719 s	3,253	204	Individual Assignment 1 sample trace

These are sample values carried forward from the Assignment 1 evidence. They are not a controlled average and should be compared with Assignment 2 only after equivalent tasks are run and measured in LangSmith.

12. Limitations and Responsible Use

- The supplied students.json explicitly labels the records as demo data. It must not be treated as production institutional data or production authentication.
- Local JSON storage is suitable for a classroom prototype, not a secure multi-user production deployment.
- The submitted ZIP includes a .env file. Remove it from any public repository or shared submission package; keep credentials private and submit .env.example instead.
- Assignment 2 latency/token measurements and LangSmith trace export were not present in the supplied ZIP. These should be added from actual runs rather than estimated.
- Attach current screenshots/video frames for the on-demand contact-details feature and the final confirm/reject messages if these were added after the ZIP was created.

13. Submission Checklist

- Updated working Student Support Agent project ZIP (exclude .env and real credentials).
- This submission report (PDF and editable DOCX).
- Final evaluation workbook (XLSX).
- Sample audit evidence (included in report and workbook).
- LangSmith trace screenshots/export showing tool calls, latency, input tokens and output tokens.
- Video demonstration link with access set so the evaluator can view it.
- Evidence screenshots for own-record access, cross-student denial, approval pending, rejection, successful confirmation, and OTP/password memory exclusion.

14. Conclusion

Assignment 2 extends the original Student Support Agent with privacy-aware access control, explicit approval for sensitive contact changes, persistent-memory safeguards, and auditability. The supplied audit and approval JSON files demonstrate the main permission and approval outcomes. Quantitative Assignment 2 performance comparison remains incomplete until current LangSmith measurements are exported and attached.